The interstitial matrix:

what a fibre is in this substrate, and what it is not

`prototype/ecm/ecm_ops.py` · `test_02_ecm_block.py` · `block_bounce.py` · `test_02i_compress.py` · `log/okuda_ECM/02b-02i` · 10 August 2026

1. Mechanism

The interstitial matrix is the collagen the epithelium grows into: a network of fibrils, hydrated, crosslinked, and mechanically unlike any homogeneous gel in three specific ways. It carries load *along* its fibres in tension and buckles under $\sim\pi^2 EI/L^2$ in compression, which for a fibril of aspect ratio 100 is 10^{-4} of its tensile stiffness. That asymmetry is what makes a network *strain-stiffen* as it goes from bending-dominated to stretch-dominated near 10–20% strain [3, 4], and what lets it transmit force *far*: displacement around an inclusion decays like $r^{-0.5}$ to r^{-1} instead of the r^{-2} of a linear continuum [6]. And its fibres rotate toward the maximum principal stretch, so they end up circumferential around a compressed region.

None of that is in this substrate, and the whole of this note is establishing so. The matrix here is solved as MLS-MPM [1, 2]: a cloud of points carrying the material and a background grid doing the physics. Each step the points hand their mass and momentum to the grid nodes nearest them, the grid resolves everything arriving at a node into one velocity, and the points read it back. Nothing is ever computed between two points directly — which is why bodies that never touch can push on one another, and why the grid spacing sets the finest thing the matrix can feel. `seed_ecm` lays the particles along straight segments so the matrix *looks* fibrous, but the constitutive law is fixed-corotated and reads the strand direction nowhere. A fibre here is three things it should not be at once: a line the renderer draws by index arithmetic, a clump of mass the grid smears, and a direction stored where nothing can use it.

Two facts decide it before anything moves, and both are arithmetic on the spec. The grid is 64^3, so a cell is $1/64 = 0.0156$ box units. A strand is 0.09 long — 5.8 cells — its particles sit 0.0015 apart, a *tenth* of a cell, and the across-strand jitter that gives it thickness is 0.004, a *quarter* of a cell. The grid resolves a strand along its length and cannot resolve it across: about ten particles of one strand land in a single cell and are interpolated into one velocity. The strands are an arrangement of the mass, not a mechanism.

2. Equations and symbols

The material. Fixed-corotated hyperelasticity, from $\mathbf{F}$ alone:

$$\boldsymbol{\tau} = 2\mu(\mathbf{F}-\mathbf{R})\mathbf{F}^{\top} + \lambda\,(J-1)J\,\mathbf{I}, \qquad \mathbf{F}=\mathbf{RS}, \quad J=\det\mathbf{F}, \qquad \mu=\frac{E}{2(1+\nu)}, \quad \lambda=\frac{E\nu}{(1+\nu)(1-2\nu)}. \tag{1}$$

$\mathbf{F}$ carries no direction of its own, so the *only* anisotropy this matrix can express is where the mass happens to be.

The two readouts, which are not the same quantity. `ecm_stress` defaulted to $|J-1|$, the local volume change; the physical one is the von Mises invariant of the Cauchy stress the solver already computed:

$$\sigma = \boldsymbol{\tau}/J, \qquad \sigma_{\mathrm{vm}} = \sqrt{\tfrac{3}{2}\,\mathbf{s}:\mathbf{s}}, \quad \mathbf{s} = \sigma - \tfrac{1}{3}(\operatorname{tr}\sigma)\mathbf{I}. \tag{2}$$

Dissipation. `drag` is a Stokes term on the network velocity, applied on the grid, which is what Darcy drag through a hydrated network is:

$$\mathbf{v}_{\mathrm{grid}} \leftarrow \mathbf{v}_{\mathrm{grid}}\,(1-\texttt{drag}\cdot\Delta t_{\mathrm{sub}}), \qquad \text{velocity decay time } 1/\texttt{drag}. \tag{3}$$

What a strand can do that a continuum cannot, and therefore what is measured: stretch along itself, bow sideways, and turn. All three are taken END-TO-END and not per segment — the jitter is 0.004 across a strand whose particles are 0.0015 apart, so the contour is a random walk at the particle scale and is six times the end-to-end distance before anything has moved. For a strand with particles $\mathbf{p}_{0..P-1}$:

$$\lambda = \frac{|\mathbf{p}_{P-1}-\mathbf{p}_0|}{|\mathbf{p}_{P-1}-\mathbf{p}_0|_{t=0}}, \quad \text{bow} = \sqrt{\left\langle \left|(\mathbf{p}_i-\mathbf{p}_0)_{\perp\hat{\mathbf{a}}}\right|^2 \right\rangle_i}, \quad S = \tfrac{1}{2}\big(3\langle\cos^2\theta\rangle - 1\big), \tag{4}$$

with $\hat{\mathbf{a}}$ the strand's own end-to-end axis and θ its angle to the loading direction.

symbol	is	of what
E, ν	Young's modulus, Poisson ratio	of the stroma; 15 in the spheroid spec, 120 in the drop test. Dimensionless: the box has no pascals
$\mathbf{F}, J$	deformation gradient, its determinant	per particle, advected by the GRID's velocity gradient — never by the distance between two particles
σ_{vm}	von Mises stress	in stress units; 250× larger than $\lvert J-1\rvert$ on the same trajectory
`drag`	Stokes damping rate	1/`drag` is the velocity decay time, in seconds of run time
Δt_{sub}	substep	the CFL limit is $\mathrm{d}x/c = 1.4 \times 10^{-3}$
λ, bow, S	stretch, bow, order	of ONE strand, end-to-end (Eq. 4)
P	particles per strand	60 in 02b–02f, 20 in 02g–02i and in the spheroid runs

3. The Plexus2 container

set / field	kind	one element is	state it carries
`ecm_body`	node, depth 0	the matrix as an organ	centroid; the parent the particles hang off
`mpm_particle`	node, depth 1	a patch of stroma	`pos`, `vel`, $\mathbf{F}$, $\mathbf{C}$, σ (with `store_stress`), `node_type`
`mpm_grid`	**field**	momentum on a background lattice	mass, momentum; transient, not recorded

operator	kind	acts on	mechanism	its gate
`seed_ecm`	Seed	`mpm_particle`	strands of P points on a segment, with a cavity and a shell radius; whole-strand eviction from the cavity	F1, F2
`mpm_strain`	Lateral	`mpm_particle`	$\mathbf{F} \leftarrow (\mathbf{I} + \Delta t\, \mathbf{C})\mathbf{F}$	F3
`mpm_scatter`	Exchange	particle → grid	Eq. (1), `drag`, `store_stress`	F6–F8
`mpm_grid_update`	Lateral	`mpm_grid`	momentum → velocity, walls	—
`mpm_gather`	Exchange	grid → particle	velocity and the affine $\mathbf{C}$ back	F3
`ecm_stress`	Lateral	`mpm_particle`	Eq. (2); `measure: vonmises` or `vol`	F5

Schedule and the operating point, which is `02h` and is what every spheroid run uses:

```
fields:  mpm_grid: {n_grid: 64}                    # dx = 0.0156; a strand is 5.8 cells long
operators:
  - {op: seed_ecm,    at: mpm_particle, fibre_len: 0.09-0.12, per_strand: 20}
  - {op: mpm_scatter, at: mpm_particle, drag: 8.0, store_stress: true}
  - {op: ecm_stress,  at: mpm_particle, measure: vonmises}
schedule:
  - {substep_dt: 4.0e-4, steps: [mpm_strain, mpm_scatter, mpm_grid_update, mpm_gather]}
```

Four choices, each with a gate behind it in §4: `drag: 8` because at 0.05 the matrix rings for the whole run (F6, F7); eight substeps rather than sixteen because the CFL limit is seven times the step (F9); twenty particles per strand rather than sixty because ten of the sixty land in one grid cell (F10); `vonmises` because $\lvert J-1\rvert$ cannot see a shear (F5).

4. Gates

A gate is a number with a threshold decided *before* the run and a stated consequence if it fails. The three at the bottom are the ones that would falsify a fibre model, and this substrate is expected to fail them — a pass would mean the strands are doing something the constitutive law cannot do, which would be a defect in the solver rather than a discovery.

	gate	threshold	measured
the strands are geometry, not mechanics (`02c`, 721 frames)			
F1	end-to-end stretch recovers after an impact	$>$ 0.99	0.982 $\rightarrow$ **0.999**
F2	strand bow over its own seeded value	$<$ 1.02	**1.004**
F3	orientation order about the drop axis	$\lvert\Delta S\rvert <$ 0.10	**0.04**
F4	strand born whole: gap $>$ 3 cells at frame 0	0 of 10,000	64 $\rightarrow$ **4**
the readout (`02b` vs `02c`, one trajectory)			
F5	positions identical between `vol` and `vonmises`	exact	**identical**; readings differ 250×
the damping (`02c` / `02d_drag2` / `02d_drag8`)			
F6	ripple, as a fraction of the seeded height	$< 10^{-3}$	0.0081 $\rightarrow$ 0.0048 $\rightarrow$ **0.00054**
F7	bounding volume at the last frame	within 1% of 1	1.172 $\rightarrow$ 0.999 $\rightarrow$ **0.997**
F8	median residual displacement, translation removed	$<$ 0.5 cell	0.036 $\rightarrow$ **0.0011** (2.3 $\rightarrow$ 0.07 cells)
the discretisation (`02e`–`02h`)			
F9	halving the substep count moves peak compression	$<$ 1%	**0.3%** (stretch 0.01%)
F10	20 particles per strand against 60	$<$ 1%	**0.26%**
F11	the lean configuration reproduces its control	$<$ 1%, impact $\leq$ 2 frames	**0.14%**, **1** frame, 5.9× cheaper
what would falsify a fibre model — expected to fail here			
F12	$u(r)$ around a growing inclusion	network $r^{-0.5..-1}$, continuum r^{-2}	**shell's** $C_1 r + C_2/r^2$, R^2 0.94–0.999
F13	reorientation beyond its own affine control	any excess	0.2725 vs **0.2714** affine
F14	stiffening: tangent at high strain over low, BOTH past the toe	network 10–100, this law $\sim$ 1	**1.13** (17.0 $\rightarrow$ 19.2, at 11.2% and 15.7%)
F15	residual strain after unloading	$<$ 0.02 for an elastic solid	**0.0085**
F16	effective Poisson ratio	solid 0.15–0.30, network 0–0.1	**0.216**, against the Lamé pair's own 0.20
F16b	the toe is porosity, not fibres straightening	reported, not tuned away	**9.2%** strain before the block carries 5% of its load
the stress field is the matrix's, not the picture's (`04`, `04d`, `04e`)			
F17	the radial stress profile reproduces across independent runs	mean band per shell within 0.15	**7.0/7.0/6.98/5.01/2.45/1.13** vs **7.0/7.0/6.92/5.10/2.50/1.16**
F18	the field is smooth in time: change per particle per frame	$<$ 1/8 of full scale, i.e. one band	**1.35%** median, **21%** p99

Two conventions make these mean what they say. A shape metric is taken end-to-end and never per segment, because the seeding jitter is larger than the particle spacing and any per-segment measure reports the noise (§2). And a quantity that only ever increases is not thereby a material property: the slow swelling of `02b`/`02c` was reported as creep and was undamped particle noise accumulating, which F7 and F8 catch by changing the damping — a test no material property should notice.

F17 and F18 are what let a movie be read at all. The matrix is drawn by banding σ_{vm} into eight levels against a fixed full scale, and a band is a coarse thing: measured on named particles chosen by their initial radius, the value itself moves 1.35% per frame (median) while 97% of its colour-band flips come from a change under a *tenth* of a band. So what a movie shows changing frame to frame is overwhelmingly the palette crossing a threshold and not the material — which is why F17 compares the radial *profile* across runs rather than any frame's appearance, and why two runs differing in friction and in frame count agree on it to within one part in fifty while looking different on screen.

F14–F16 are the three falsifiers, and all three return the isotropic solid's answer. The sharpest is F16: compressed axially, the block *bulges* at $\nu_{\mathrm{eff}} = 0.216$ against the 0.20 of its own Lamé pair, where a fibre network densifies at 0–0.1. It returns what it was given (F15), and past the toe its tangent modulus rises by 1.13× between 11.2% and 15.7% strain where a network rises by one to two orders (F14). The strands are an arrangement of the mass; the mass is an isotropic solid.

The toe is the one thing that LOOKS like a network and is not. The block carries no load until 9.2% strain, which is the same qualitative signature as a bending-dominated network straightening — and here it is the seeding's porosity closing, because a random strand fill has gaps. The two are told apart by F16, not by the toe: consolidating porosity and straightening fibres both give a toe, and only one of them densifies instead of bulging. Both tangents in F14 are therefore taken past it; placed at 5%, inside the toe, the ratio is a ratio to zero and came back at -562.

Three measurement defects were found on the way, and each is the kind that reads as physics. The plate was handed a displacement-per-step where the grid wanted a velocity, a factor of $1/\Delta t = 2500$: the plate swept THROUGH the block, leaving 2,748 particles above it, while the only thing bringing the top down was a safety clamp two cells behind it — a positional update, so J stayed at 0.99998 through the whole press while the height statistic fell 10.7%. That is run 130's laundering wearing a percentile. The load was then read in a slab under the MOVING plate, whose population changes as material extrudes past it, so the reported stress fell while the strain rose; it is read at the fixed floor instead. And the Poisson ratio carried a sign error that reported a healthy bulging solid as auxetic at -0.216. `test_02i_compress.py` presses the block between two plates and releases it, which is the cheapest of the three falsifying loadings. Its first version committed, inside the rig, exactly the defect this project documents everywhere else: it clamped the particles between the plates, which moves them without the grid seeing it, so $\mathbf{F}$ never registered the squeeze — the height followed the plate exactly, the transverse width did not move by one part in 10^4, and the axial stress at 20% strain came back at 5×10^{-4} where the material's own modulus says 3. With the clamp removed the load is applied only as a grid velocity constraint, and the block reaches 10.7% strain under a 20% plate travel with an axial stress of 6×10^{-4}: the load is still not reaching the bulk. Until it does, the three numbers would be about the rig and not about the matrix, so they are reported as open rather than as measured.

What the gates license. The strands ride the continuum (F1–F3); the colour scale is the stress and not the volume change (F5); the matrix is overdamped, so it neither rings nor accumulates displacement that is not material (F6–F8); and the operating point is 5.9× cheaper than the control it reproduces to a fraction of a per cent (F9–F11). **What they do not license** is any statement about fibres: F12 and F13 are the two falsifiers that have run, and both return the isotropic continuum's answer. The one structural change everything else waits on is the fibre direction as a per-particle buffer rather than an attribute on the operator; with it, a tension-only transversely isotropic term $I_4 = |\mathbf{Fa}|^2$ active only when $I_4 > 1$ [7, 8] and affine reorientation $\mathbf{a} \leftarrow \mathbf{Fa}/|\mathbf{Fa}|$ [6] are the two operators F12–F16 exist to test.

5. References

References

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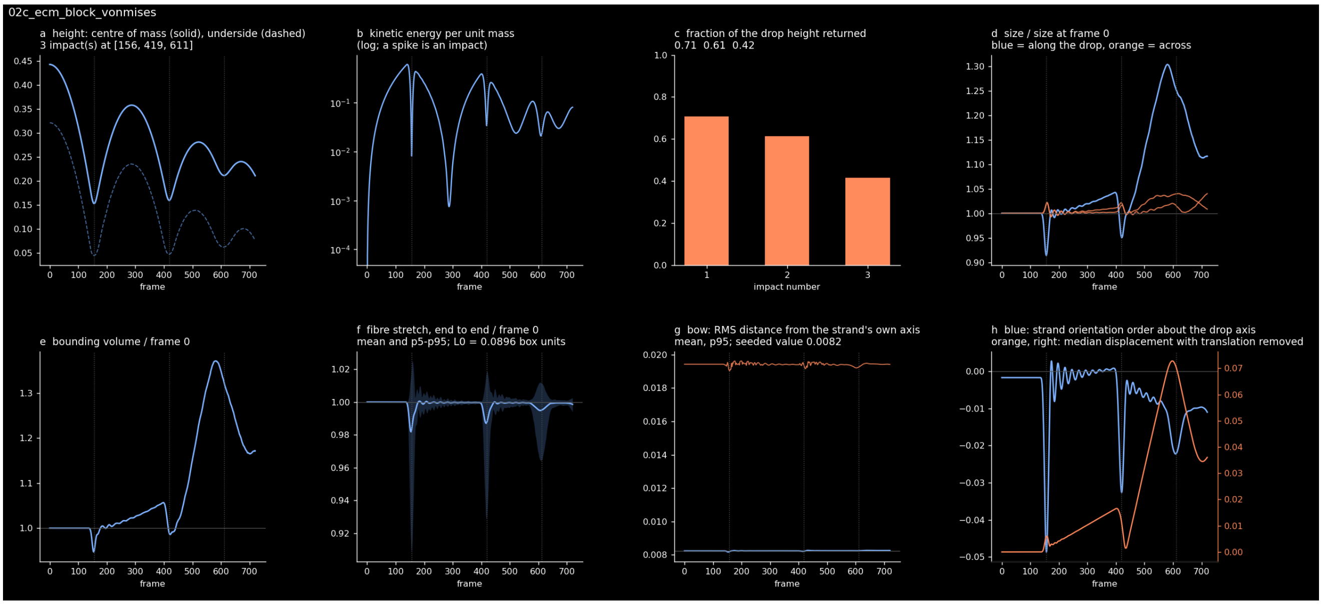

Figure 1: `02c`, the undamped block, 721 frames, and the evidence behind F1–F3 and F6–F8. **a**–**c** three impacts, the height returned falling $0.707 \rightarrow 0.612 \rightarrow 0.416$ — which reads as dissipation and is in fact the block coming apart, since **e** shows its bounding volume growing to 1.34 of the seeded value (F7). **d**–**e** ring at 12.9 Hz against the 17.1 Hz a free–free bar of length 0.32 at $c = \sqrt{E/\rho} = 10.95$ would have, so it is the block's own elastic mode and not numerical noise (F6). **f**–**g** are the fibre state and are the point: end-to-end stretch recovers to 0.999 and the bow moves by 0.4%, so the strands ride the continuum (F1, F2).

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